Report: Single-Area OSPF Configuration and Link-Failure Rerouting

Tool: Cisco Packet Tracer | **Topic:** OSPF (process 1, area 0), neighbor verification, convergence and failover

1. Objective

Three routers were configured with single-area OSPF so every LAN could reach every other LAN. The goal was to verify neighbor adjacencies and routing-table convergence, then deliberately break one link and document how traffic reroutes.

2. Topology

The three routers form a triangle, so each pair has a direct link and a backup path exists through the third router. Each router serves one LAN through a switch with two PCs.

```
            Router0 ============ Router2
               \\                 //
                \\               //
                 \\             //
                     Router1

LAN0 (PC0, PC1) on Router0   LAN1 (PC2, PC3) on Router1   LAN2 (PC4, PC5) on
Router2
```

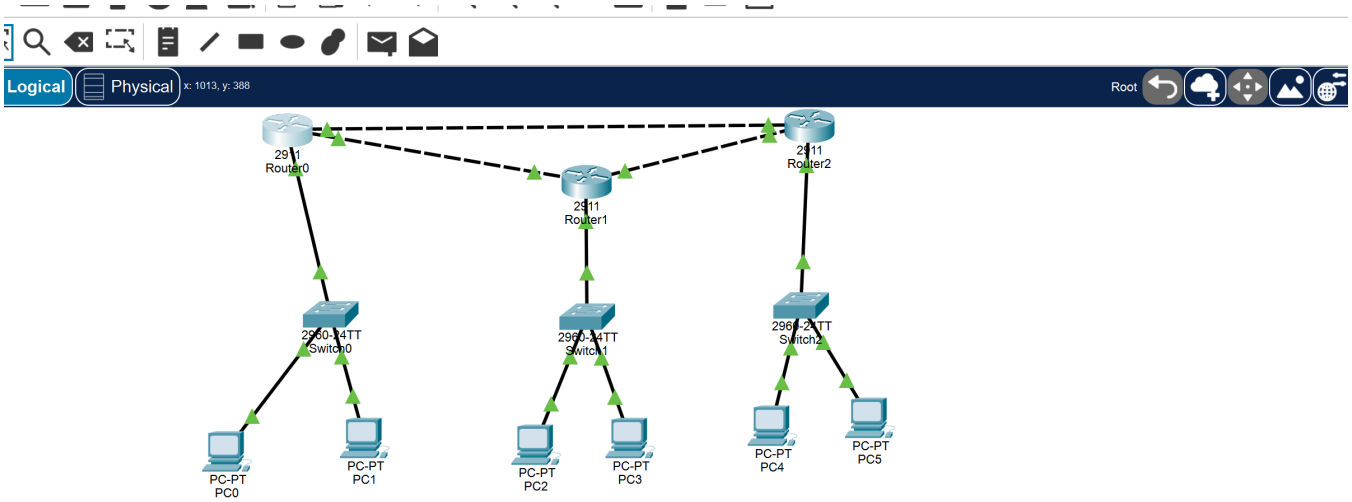


Figure 1: Packet Tracer topology, three 2911 routers in a triangle, each with a 2960 switch and two PCs.

3. Addressing Plan

Network	Subnet	Router interfaces
LAN0	192.168.1.0/24	Router0 Gi0/0 = 192.168.1.1
LAN1	192.168.2.0/24	Router1 Gi0/0 = 192.168.2.1
LAN2	192.168.3.0/24	Router2 Gi0/0 = 192.168.3.1
Router0 to Router1	10.0.12.0/30	Router0 Gi0/1 = .1, Router1 Gi0/1 = .2
Router1 to Router2	10.0.23.0/30	Router1 Gi0/2 = .1, Router2 Gi0/1 = .2
Router0 to Router2	10.0.13.0/30	Router0 Gi0/2 = .1, Router2 Gi0/2 = .2

Router	OSPF router ID
Router0	1.1.1.1
Router1	2.2.2.2
Router2	3.3.3.3

PCs use .10 and .11 in their LAN, mask 255.255.255.0, with the router's Gi0/0 address as default gateway.

4. Configuration

Each router advertises its LAN and its two point-to-point links in area 0. The LAN interface is passive so no OSPF hellos are sent toward the PCs.

Router0

```
hostname R0
interface g0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
interface g0/1
 ip address 10.0.12.1 255.255.255.252
 no shutdown
interface g0/2
 ip address 10.0.13.1 255.255.255.252
 no shutdown
router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.13.0 0.0.0.3 area 0
 passive-interface g0/0
```

Router1

```
hostname R1
interface g0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
interface g0/1
 ip address 10.0.12.2 255.255.255.252
 no shutdown
interface g0/2
 ip address 10.0.23.1 255.255.255.252
 no shutdown
router ospf 1
 router-id 2.2.2.2
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.23.0 0.0.0.3 area 0
 passive-interface g0/0
```

Router2

```
hostname R2
interface g0/0
 ip address 192.168.3.1 255.255.255.0
 no shutdown
interface g0/1
 ip address 10.0.23.2 255.255.255.252
 no shutdown
interface g0/2
 ip address 10.0.13.2 255.255.255.252
 no shutdown
router ospf 1
 router-id 3.3.3.3
 network 192.168.3.0 0.0.0.255 area 0
 network 10.0.23.0 0.0.0.3 area 0
 network 10.0.13.0 0.0.0.3 area 0
 passive-interface g0/0
```

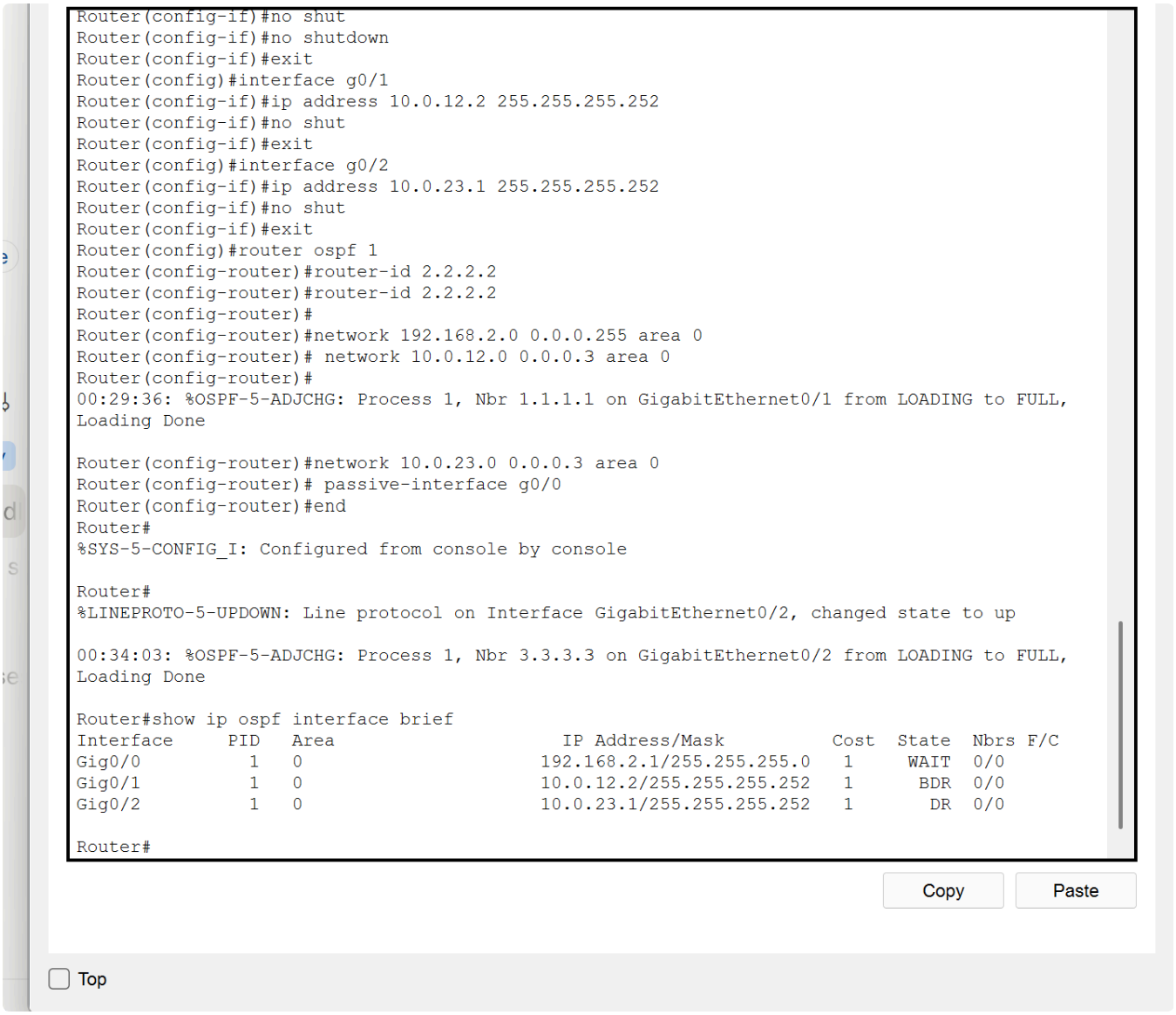


Figure 2: Router1 CLI session showing the OSPF configuration and `show ip ospf interface brief`. All three interfaces are in area 0 with cost 1. The state column (WAIT, BDR, DR) was captured while adjacencies were still forming, so Nbrs reads 0/0; the neighbor table below shows them FULL.

5. Verification Before the Failure

5.1 Router1: neighbors and learned routes

Router1 has two FULL adjacencies and learned both remote LANs through OSPF.

Neighbor ID	State	Address	Interface
1.1.1.1 (Router0)	FULL/DR	10.0.12.1	GigabitEthernet0/1
3.3.3.3 (Router2)	FULL/BDR	10.0.23.2	GigabitEthernet0/2

Code	Route	Metric	Next hop	Interface
O	192.168.1.0/24	[110/2]	10.0.12.1	GigabitEthernet0/1
O	192.168.3.0/24	[110/2]	10.0.23.2	GigabitEthernet0/2

The `O` code marks OSPF-learned routes. Cost 2 equals the link cost (1) plus the destination LAN's interface cost (1). Both remote LANs are reached directly over a single link, which confirms the routing table converged.

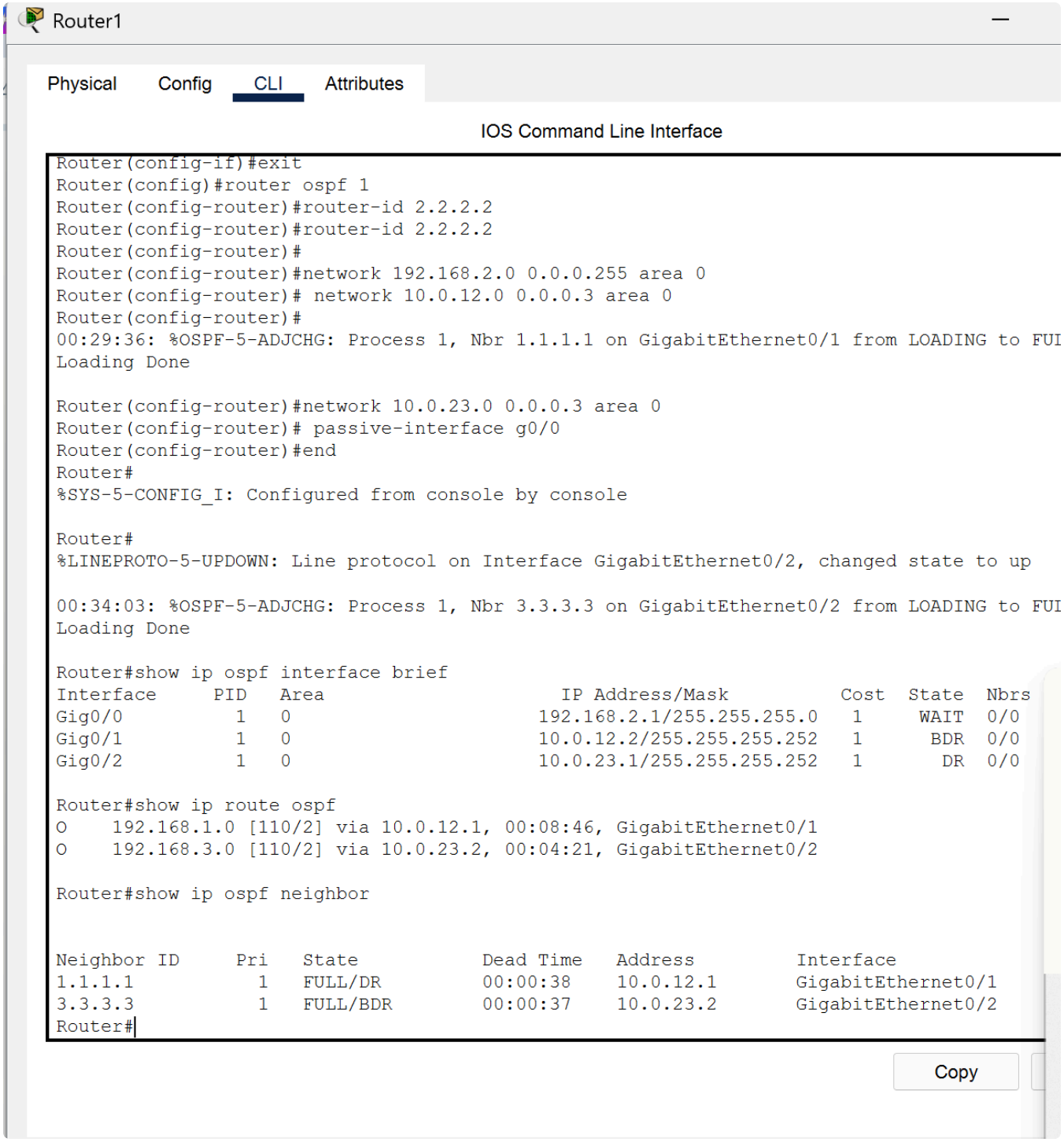


Figure 3: Router1 `show ip route ospf` and `show ip ospf neighbor`.

5.2 Router0: neighbor

Router0's `show ip ospf neighbor` in this capture lists Router1 (2.2.2.2) in state FULL/BDR on Gi0/1 via 10.0.12.2. The Router0 to Router2 link had not yet come up when this was captured, so only one neighbor appears.

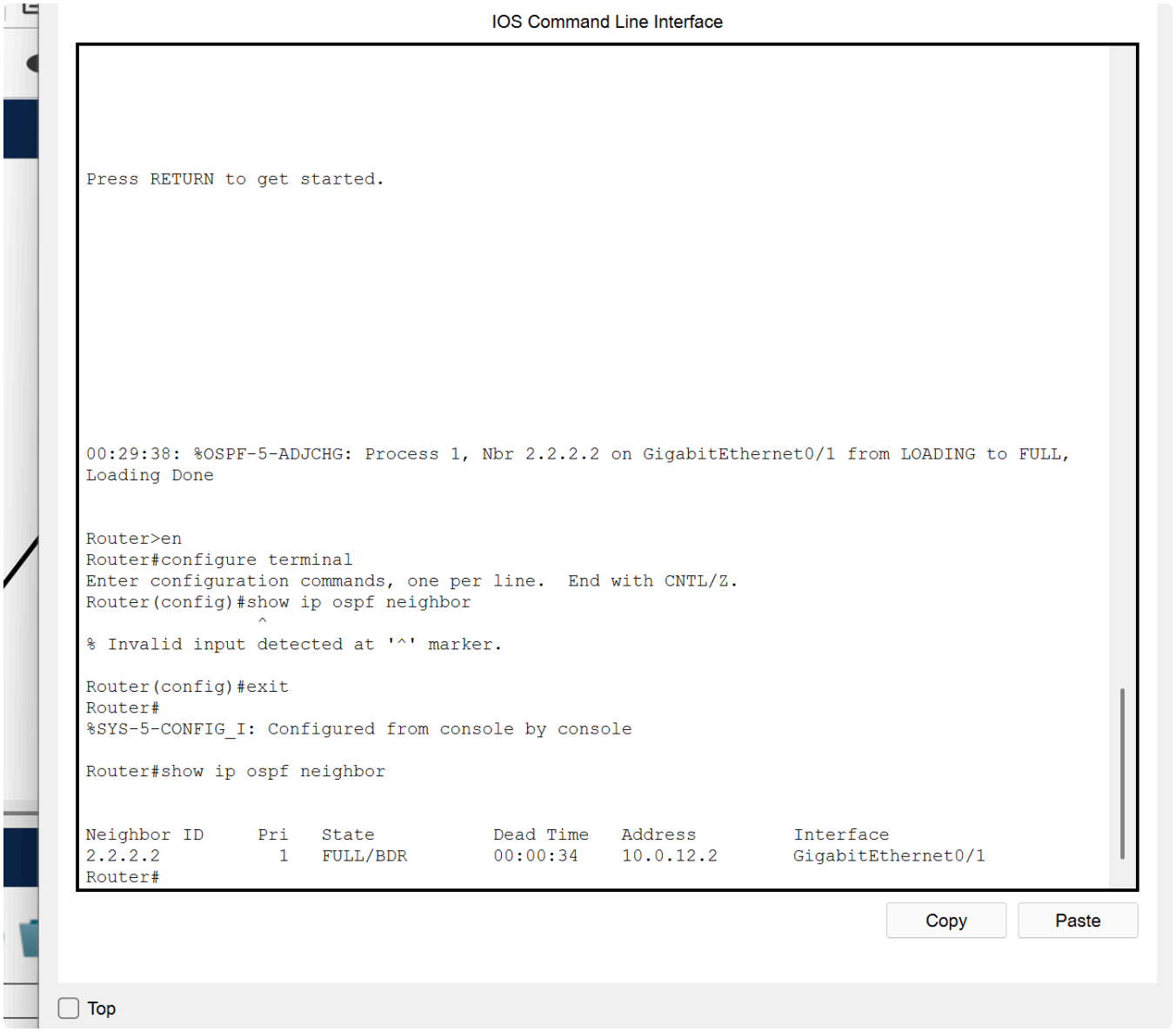


Figure 4: Router0 `show ip ospf neighbor` before the failure.

5.3 End-to-end test (PC0 to PC2)

PC0 pinged 192.168.2.10 twice. The first attempt lost all 4 packets and the second lost 1 of 4 (25% loss), which is typical while ARP entries and OSPF adjacencies are still settling. The replies that did arrive had TTL 126, meaning the packets crossed two routers.

```
1   192.168.1.1     (Router0, LAN0 gateway)
2   10.0.12.2       (Router1, link interface)
3   192.168.2.10    (PC2)
Trace complete.
```

The traceroute shows the normal path is the direct one: **PC0 → Router0 → Router1 → PC2**.

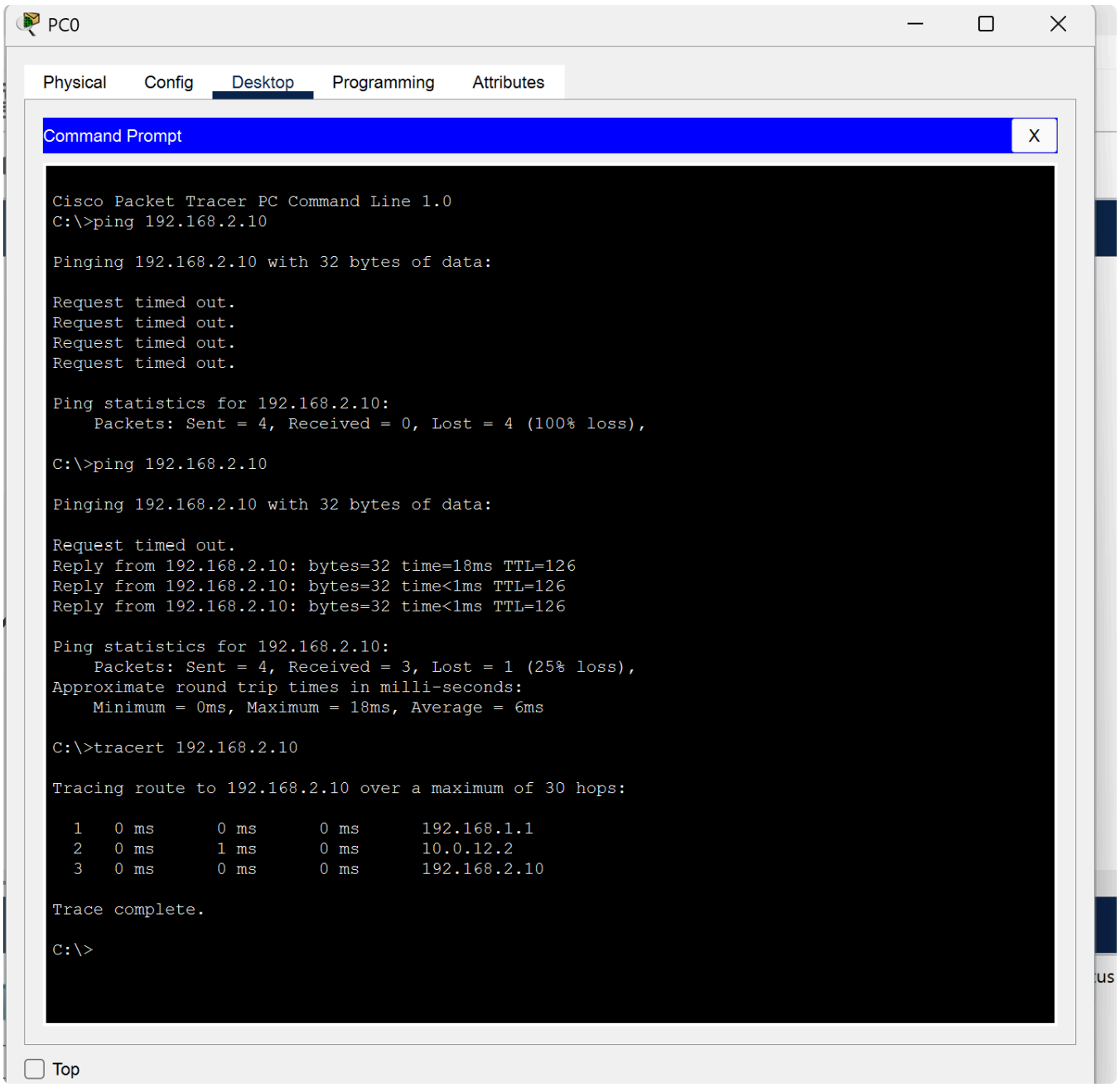


Figure 5: PC0 ping and tracert to 192.168.2.10 before the failure.

6. Breaking the Link

The Router0 to Router1 link was brought down by shutting Router0's Gi0/1.

```
R0(config)# interface gi0/1
R0(config-if)# shutdown
```

Packet Tracer immediately logged the loss of the adjacency:

```
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
administratively down
%OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from FULL to
DOWN, Neighbor Down: Interface down or detached
```

Router0's `show ip ospf neighbor` afterwards no longer lists 2.2.2.2. Only 3.3.3.3 (Router2) remains, reached through Gi0/2 at 10.0.13.2. It shows 2WAY/DROTHER with a dead time of 00:00:34, because the capture was taken just as Gi0/2's line protocol came up and the adjacency was still negotiating.

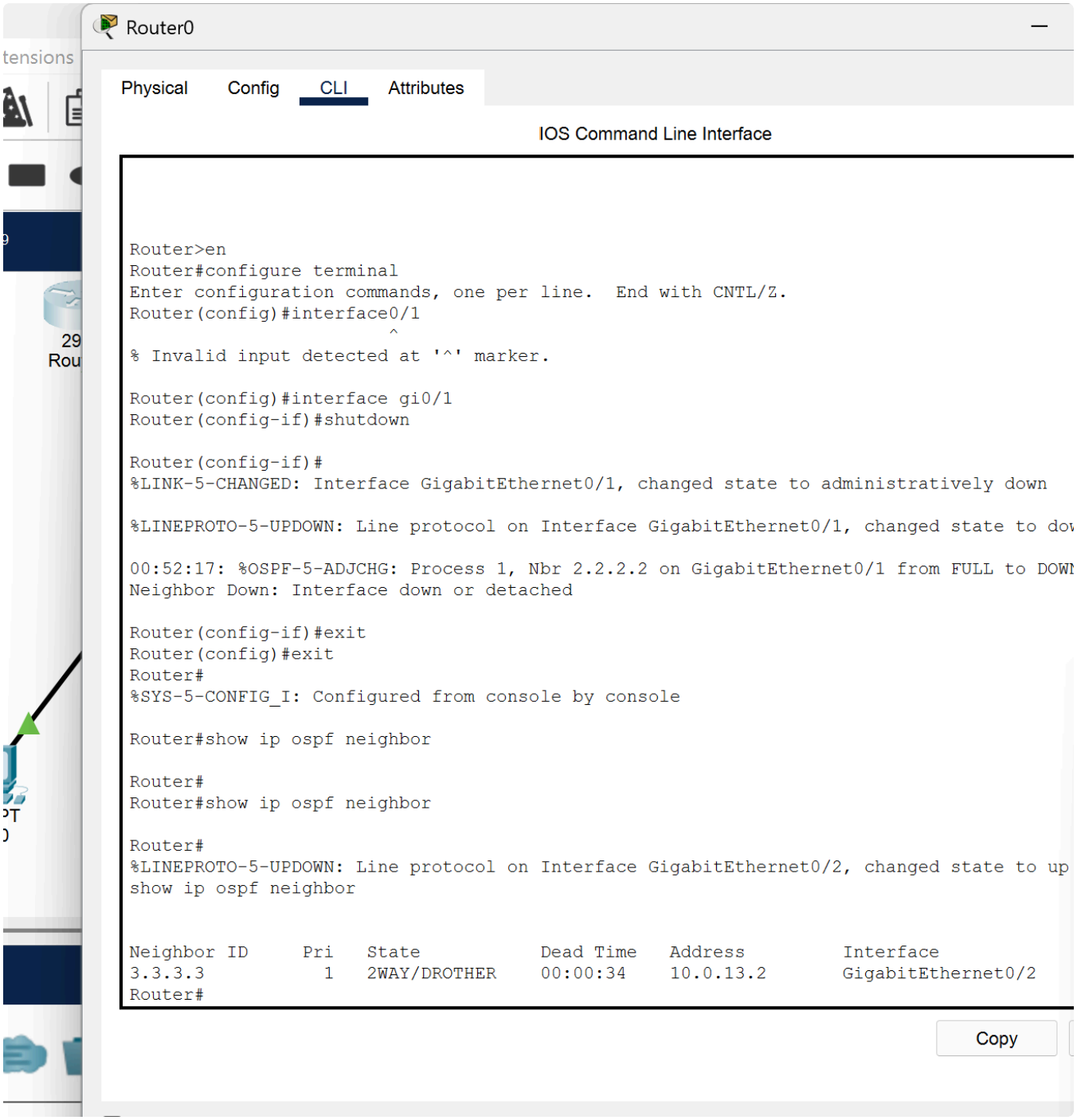


Figure 6: Router0 shutting Gi0/1, the neighbor-down log message, and `show ip ospf neighbor` after the failure.

7. Before and After: How Traffic Reroutes

The table compares Router0's path to LAN1 (192.168.2.0/24). Costs follow from the interface costs shown in Figure 2 (1 per Gigabit link). The "after" values are the expected result of the topology and should be confirmed with a `show ip route ospf` capture on

Router0.

	Before failure	After failure
Neighbors on Router0	2.2.2.2 and 3.3.3.3	3.3.3.3 only
Route to 192.168.2.0/24	via 10.0.12.2 (Gi0/1), cost 2	via 10.0.13.2 (Gi0/2), cost 3
PC0 to PC2 path	Router0 → Router1	Router0 → Router2 → Router1
Expected tracert hops	192.168.1.1, 10.0.12.2, 192.168.2.10	192.168.1.1, 10.0.13.2, 10.0.23.1, 192.168.2.10

When the link failed, Router0 removed the Router1 adjacency, OSPF flooded updated LSAs, every router re-ran SPF, and the route to LAN1 moved to the longer path through Router2. No manual change was made.

8. Results

Test	Outcome
All routers in OSPF process 1, area 0	Done
Router1 adjacencies FULL with both neighbors	Passed
Router1 learned both remote LANs as `O` routes	Passed
PC0 to PC2 reachable, path via Router1	Passed
Link failure detected (FULL to DOWN logged)	Passed
Rerouted path via Router2 shown with routing table and tracert	Needs screenshot (see section 10)

9. Conclusion

Single-area OSPF gave all three LANs full reachability without any static routes, and the routing table showed OSPF-learned `O` entries with the expected costs. Shutting one link removed that adjacency within moments, and the triangle topology left a second path

through Router2 for traffic to take. A plain chain of routers would have had no such backup path.

Advantages of OSPF: automatic convergence, fast failure detection, cost-based best-path selection, no manual route updates. **Considerations:** more configuration and CPU than static routing; a single area does not scale to very large networks.

10. Screenshots Still Needed for a Complete Submission

1. `show ip route ospf` on Router0 before the failure
2. `show ip route ospf` on Router0 after the failure (should show 192.168.2.0/24 via 10.0.13.2, cost 3)
3. `tracert 192.168.2.10` from PC0 after the failure
4. `show running-config` on Router0, Router1 and Router2
5. `show ip ospf neighbor` on Router0 after waiting about 40 seconds, so Router2 shows FULL